.1 development of accident scenarios;

.2 probability of events in each accident scenario; and

.3 consequence of end states in each scenario;

.3 assess risks and determine acceptability:

.1 estimate risk levels in accordance with the selected modelling approach;

.2 assess whether risk levels are acceptable; and

Reference is made to the techniques in appendix 3 of the Revised guidelines for Formal Safety Assessment (FSA) for use in the IMO rule-making process (MSC-MEPC.2/Circ.12) and standard IEC/ISO 31010 "Risk management Risk assessment techniques".
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"Risk management Risk assessment techniques".
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.4 in the event that risk levels determined in steps 1 to 3 are considered to be too high, identify current or develop new risk control options that aim to achieve one or more of the following:

.1 reduce the frequency of failures through better design, procedures, training, etc.;

.2 mitigate the effect of failures in order to prevent accidents;

.3 limit the circumstances in which failures may occur; or

.4 mitigate consequences of accidents; and

.5 incorporate risk control options for design, procedures, training and limitations, as applicable.

2.3 Performance standards

A system previously accepted based on manufacturer certifications, classification society certifications and/or satisfactory service of existing systems may be acceptable for installation on new and existing ships if no performance or testing standards are accepted by the Organization.

3 Additional guidance to chapter 2 (Polar Water Operational Manual (PWOM))

3.1 Recommendation on the content of the PWOM

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installation on new and existing ships if no performance or testing standards are accepted by the Organization.

3 Additional guidance to chapter 2 (Polar Water Operational Manual (PWOM))

3.1 Recommendation on the content of the PWOM

3.1.1 The PWOM is intended to address all aspects of operations addressed by chapter 2 of part I-A. When appropriate information, procedures or plans exist elsewhere in a ship 's documentation, the PWOM itself does not need to repli cate this material, but may instead cross-reference the relevant reference document.

3.1.2 A model table of contents is set out in appendix II.

3.1.3 The model follows the general structure of chapter 2. Not every section outlined below will be applicable to every polar ship. Many category C ships that undertake occasional or limit polar voyages will not need to have procedures for situations with a very low probability of occurrence. However, it may still be advisable to retain a common structure for the PWOM as a reminder that if assumptions change then the contents of the manual may also need to be updated . Noting an aspect as "not applicable " also indicates to the

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for the PWOM as a reminder that if assumptions change then the contents of the manual may also need to be updated . Noting an aspect as "not applicable " also indicates to the Administration that this aspect has been considered and not merely omitted.

3.2 Guidance on navigation with icebreaker assistance

With respect to navigation with icebreaker assistance, the following should be considered:

.1 while approaching the starting point of the ice convoy to follow an icebreaker/icebreakers or in the case of escorting by icebreaker of one ship to the point of meeting with the icebreaker, ships should establish radio communication on the VHF channel 16 and act in compliance with the icebreaker's instructions;

.2 the icebreaker rendering the icebreaker assistance of ship ice convoy should command ships in the ice convoy;

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.3 position of a ship in the ice convoy should be determined by the icebreaker rendering the assistance;

.4 ship within the ice convoy, in accordance with the instructions of the

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.3 position of a ship in the ice convoy should be determined by the icebreaker rendering the assistance;

.4 ship within the ice convoy, in accordance with the instructions of the icebreaker rendering the assistance, should establish communication with the icebreaker by VHF channel indicated by the icebreaker;

.5 the ship, while navigating in the ice convoy, should ensure compliance with the instructions of the icebreaker;

.6 position in the ice convoy, speed and distance to a ship ahead should be as instructed by the icebreaker;

.7 the ship should immediately notify the icebreaker of any difficulties to maintain the position within the ice convoy, speed and/or distance to any other ship in the ice convoy; and

.8 the ship should immediately report to the icebreaker of any damage.

3.3 Guidance on the development of contingency plans

In developing the ship's contingency plans ships should consider damage control measures arrangements for emergency transfer of liquids and access to tanks and spaces during salvage operations (see also additional guidance to chapter 9).

4 Additional guidance to chapter 3 (Ship structure)

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arrangements for emergency transfer of liquids and access to tanks and spaces during salvage operations (see also additional guidance to chapter 9).

4 Additional guidance to chapter 3 (Ship structure)

Method for determining equivalent ice class

4.1 The guidance presented below is intended to assist in determining equivalency with standards acceptable to the Organization, as referenced in chapters 3 and 6 of the Code. The methodology is consistent with guidance developed by the Organization while allowing for the use of a simplified approach.

4.2 The basic approach for considering equivalency for categories A and B ships can be the same for both new and existing ships. It involves comparing other ice classes to the IACS Polar Classes. For ice classes under category C, additional information on comparisons of strengthening levels is available for the guidance of owners and Administrations . The responsibility for generating the equivalency request and supporting information required should rest with the owner/operator. Review/approval of any equivalency request should be

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responsibility for generating the equivalency request and supporting information required should rest with the owner/operator. Review/approval of any equivalency request should be undertaken by the flag State Administration, or by a recognized organization acting on its behalf under the provisions of the Code for Recognized Organizations (RO Code). Several classification societies have developed easy- to-use tools for determination of compliance with the IACS Polar Class structural requirements, as have some Administrations and other third parties.

4.3 The scope of a simplified equivalency assessment (see paragraph 4.6 below) is expected to be limited to materials selection, structural strength of the hull and propulsion machinery.

Refer to the Guidelines for the approval of alternatives and equivalents as provided for in various IMO instruments (MSC.1/Circ.1455).

Refer to the annex to HELCOM Recommendation 25/7, Safety of Winter Navigation in the Baltic Sea Area, available at www.helcom.fi

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